



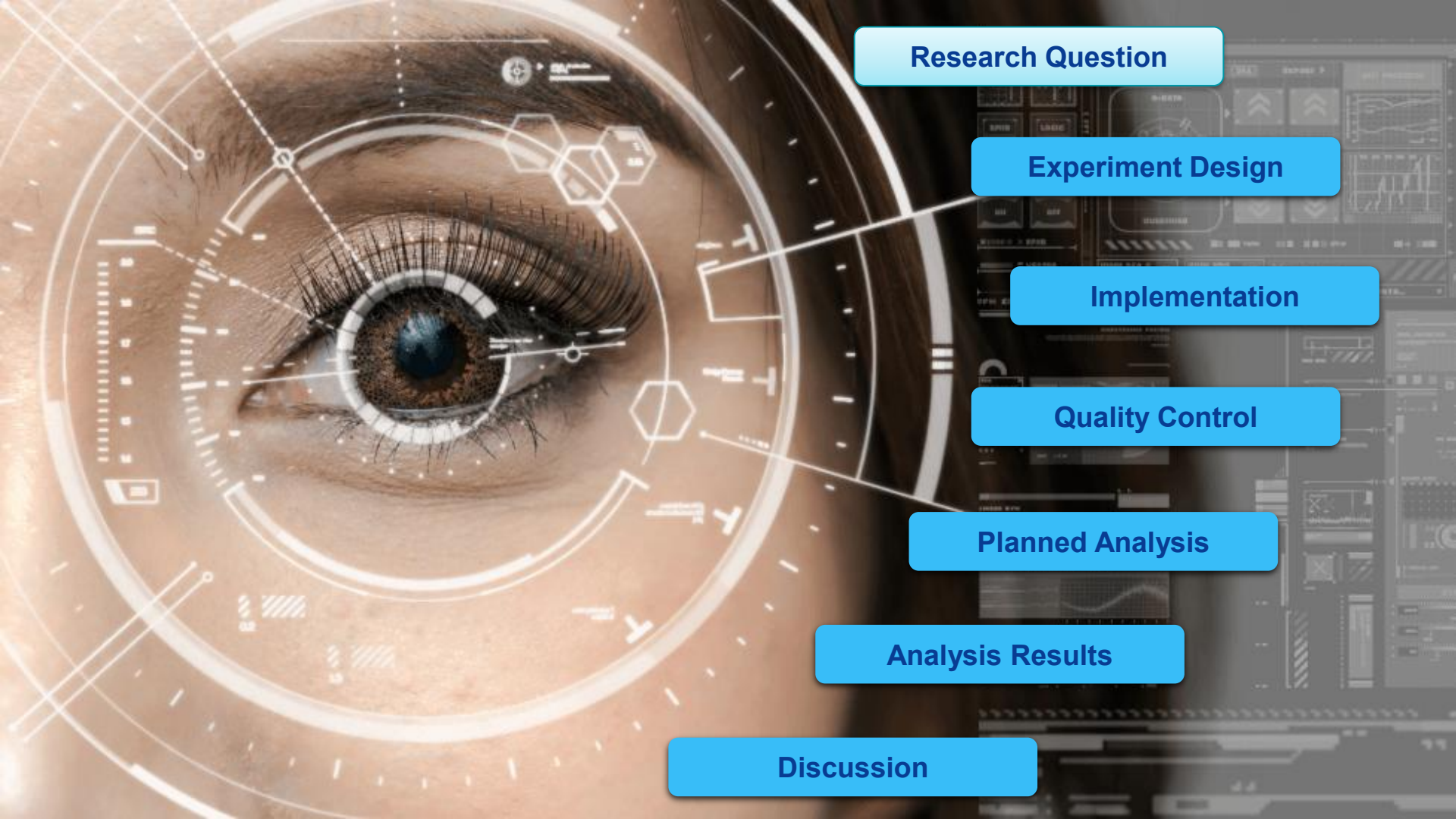
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Implicit gender bias in language processing

**Acquisition and analysis of
eye-tracking data**



Research Question

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Background & Motivation

Why does grammatical gender matter?

German role nouns can appear in different forms:

- **Masculine**: Studenten
- **Feminine**: Studentinnen
- **Gender-inclusive**: Student*innen

Previous research suggests:

- Different forms influence readers mental representations (*Redl et al., 2021*)
- Masculine forms evoke stronger male bias (*Irmen 2007; Körner et al., 2022*)
- Most effects appear during language processing (*Redl et al., 2021*)

Research Question

Do different forms of German role nouns (masculine, feminine, and gender-inclusive) elicit male and female biases in native German speakers, and what is the magnitude of these biases?

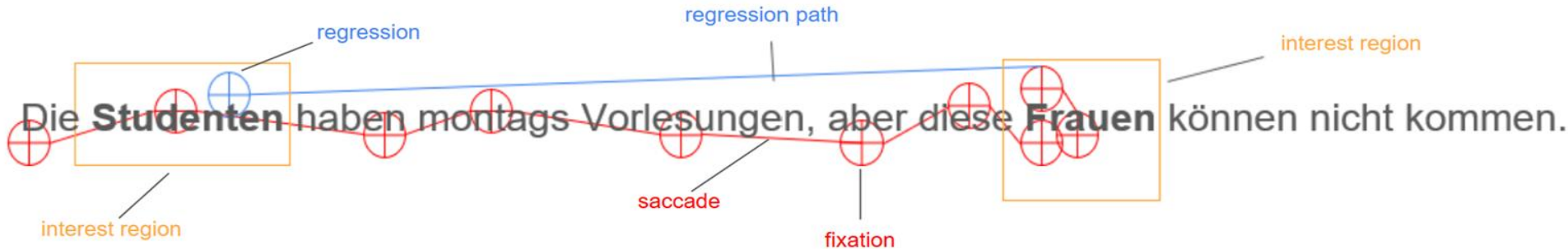
Hypotheses

- **H1:** Masculine forms produce the **strongest male bias**.
- **H2:** Feminine forms produce the **strongest female bias**.
- **H3:** Gender-inclusive forms produce the **weakest overall gender bias**.

Experiment Idea

- Participants read German sentences
- Comparison for how different forms influence readers gender expectations
- Measure both the direction (male vs. female bias) and the strength of the bias

Stimuli Example:



Note: **Masculine form**: Studenten; **Female form**: Studentinnen; **Star form**: Student*innen;



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Experiment Design

3 x 2 within-subjects design: role noun form (masculine, feminine, gender-inclusive) x anaphor type (Männer vs. Frauen)

- Results in 6 experimental conditions

Core logic: tests whether a mismatch between role noun's implied gender and the following anaphor causes increased processing difficulty

Masculine	Männer
Masculine	Frauen
Feminine	Männer
Feminine	Frauen
gender-inclusive	Männer
gender-inclusive	Frauen

Stimulus Material and Trial Procedure

108 sentences constructed **across all conditions**

Each participant reads only 18 sentences

Sentences distributed across lists so every participant sees all 6 conditions exactly 3 times

Fixed sequence per trial:

- Instruction (only for the first sequence)
- Fixation Cross
- Sentence presentation
- Comprehension question



- ✓ Masking the Research Hypothesis
- ✓ Reducing Pattern Recognition Across Trials
- ✓ Preserving Internal Validity

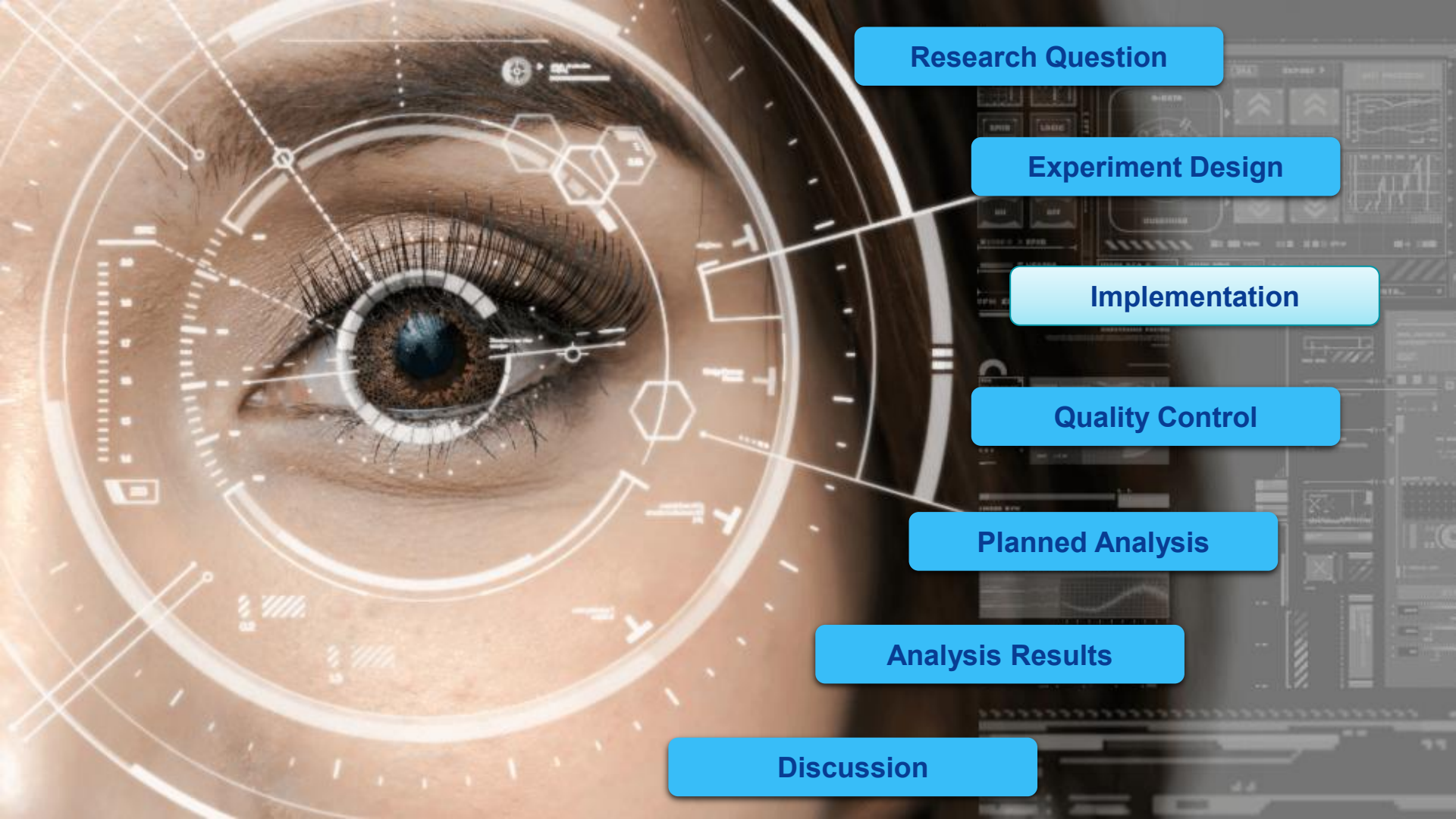
Rationale for within-subject design

Managing Order and Carry-Over Effects

- managing order effects -> practice, fatigue
- managing carry-over from earlier trials influencing later ones
- This list-based counterbalancing approximates a **Latin square logic**

Controlling Individual Reading Behavior

- Eye movement patterns during reading vary between individuals due to **reading skill**, **working memory capacity**, and general **processing speed**
- within each participant, each condition is counterbalanced



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Instruction

In jedem Durchgang erscheint zunächst ein Fixationskreuz. Anschließend wird ein Text angezeigt.

Bitte lesen Sie jeden Text aufmerksam und in Ihrem natürlichen Lesetempo.
Wenn Sie den Text vollständig gelesen haben, drücken Sie eine Taste, um zum nächsten Bildschirm zu gelangen.

Danach wird eine kurze Verständnisfrage gestellt.

Fixation Cross



Sentences

Im Rahmen einer Geschäftsveranstaltung findet derzeit ein Austausch über Arbeitsbedingungen und gesellschaftliche Entwicklungen statt. Die Arbeitnehmer*innen erwarten höhere Gehälter und klimatisierte Arbeitsplätze, aber diese Männer wollen auch weniger arbeiten. Die Frage nach den Konsequenzen bleibt offen.

Comprehension Question

Um welches Thema ging es in der vorherigen Situation?

Lists

- Lists **assigned manually** to ensure equal amount of presentations

```
1# subject gets list 1
2var.selected_list = 1
```

trial_sequence		
	list_filter	True Always
	sentence_script	[skip_trial] = 0
	sentence_pygaze_log	[skip_trial] = 0
	question_sketchpad	[skip_trial] = 0
	question_pygaze_log	[skip_trial] = 0

```
1if int(var.list) != int(var.selected_list):
2    var.skip_trial = 1
3else:
4    var.skip_trial = 0
5
```

trial_loop – loop
Führt wiederholt ein anderes Element aus

Ausführen trial_sequence Abbrechen, wenn ☐ False Never break

Wiederholen ^v ☒ Bewerten im ersten Zyklus

Reihenfolge random ☐ Nach der Pause fortsetzen

Quelle Vollfaktorielles Design

Zusammenfassung: trial_sequence wird 108 times in random Reihenfolge aufgerufen. Die Anzahl der Zeilen beträgt 108. Alle Zeilen treten einmal auf.

	item	list	sentence_text	RN	AN	Code
1	1	1	Bei einer Podiumsdiskussion an der Universität wird über aktuelle gesellschaftliche Themen gesprochen. ...	<RN> Studenten</RN>	<AN> Männer</AN>	C1
2	1	1	Bei einer Podiumsdiskussion an der Universität wird über aktuelle gesellschaftliche Themen gesprochen. ...	<RN> Studenten</RN>	<AN> Frauen</AN>	C2
3	1	1	Bei einer Podiumsdiskussion an der Universität wird über aktuelle gesellschaftliche Themen gesprochen. ...	<RN> Studentinnen</RN>	<AN> Männer</AN>	C3
4	1	1	Bei einer Podiumsdiskussion an der Universität wird über aktuelle gesellschaftliche Themen gesprochen. ...	<RN> Studentinnen</RN>	<AN> Frauen</AN>	C4
5	1	1	Bei einer Podiumsdiskussion an der Universität wird über aktuelle gesellschaftliche Themen gesprochen. ...	<RN> Student*innen</RN>	<AN> Männer</AN>	C5
6	1	1	Bei einer Podiumsdiskussion an der Universität wird über aktuelle gesellschaftliche Themen gesprochen. ...	<RN> Student*innen</RN>	<AN> Frauen</AN>	C6
7	2	1	Im Rahmen einer Gesprächsrunde im Gesundheitswesen wird derzeit über aktuelle ...	<RN> Pfleger</RN>	<AN> Frauen</AN>	C2
8	2	1	Im Rahmen einer Gesprächsrunde im Gesundheitswesen wird derzeit über aktuelle ...	<RN> Pflegerinnen</RN>	<AN> Männer</AN>	C3
100	17	4	Im Rahmen einer städtischen Planungsrunde wird über die zukünftige Mobilität und den Ausbau des ...	<RN> Ingenieure</RN>	<AN> Männer</AN>	C2
101	17	5	Im Rahmen einer städtischen Planungsrunde wird über die zukünftige Mobilität und den Ausbau des ...	<RN> Ingenieure</RN>	<AN> Frauen</AN>	C3
102	17	6	Im Rahmen einer städtischen Planungsrunde wird über die zukünftige Mobilität und den Ausbau des ...	<RN> Ingenieurinnen</RN>	<AN> Männer</AN>	C4
103	18	1	Bei einer öffentlichen Sitzung im Rathaus wird über neue Verkehrsregelungen gesprochen. Die ...	<RN> Polizist*innen</RN>	<AN> Frauen</AN>	C6
104	18	2	Bei einer öffentlichen Sitzung im Rathaus wird über neue Verkehrsregelungen gesprochen. Die ...	<RN> Polizisten</RN>	<AN> Männer</AN>	C1
105	18	3	Bei einer öffentlichen Sitzung im Rathaus wird über neue Verkehrsregelungen gesprochen. Die ...	<RN> Polizisten</RN>	<AN> Frauen</AN>	C2
106	18	4	Bei einer öffentlichen Sitzung im Rathaus wird über neue Verkehrsregelungen gesprochen. Die ...	<RN> Polizistinnen</RN>	<AN> Männer</AN>	C3
107	18	5	Bei einer öffentlichen Sitzung im Rathaus wird über neue Verkehrsregelungen gesprochen. Die ...	<RN> Polizistinnen</RN>	<AN> Frauen</AN>	C4
108	18	6	Bei einer öffentlichen Sitzung im Rathaus wird über neue Verkehrsregelungen gesprochen. Die ...	<RN> Polizist*innen</RN>	<AN> Männer</AN>	C5

Challenges – Calibration

- **Bad Calibration** can make it hard to determine whether the subject looked at the AOIs



```
# Timeout after 10 seconds
if clock.time() - t0 > 10000:
    eyetracker.log("fixation_timeout")
    eyetracker.calibrate()
    eyetracker.log("recalibration_fixation")

    fixation_canvas.show()

    t0 = clock.time()

    continue
if key == 'p':
    eyetracker.calibrate()

    c.show()

    eyetracker.log("recalibration")
    #sleep timer so no multiple calibration
    clock.sleep(200)

    continue
break
```

How can data distortion caused by poor calibration be minimized?



Recalibration after **timeout** for the fixation cross



Recalibration on **keypress**

Challenges – Skipping Words

- Words like „Männer“ and „Frauen“ appear very **frequently**
- Readers **skip** frequently occurring words

```
spill_box = {
  "x1": an_box["x2"],
  "y1": an_box["y1"],
  "x2": an_box["x2"] + an_box["width"],
  "y2": an_box["y2"],
  "width": an_box["width"],
  "height": an_box["height"],
  "line": an_box["line"]
}
```

How can we detect whether the subject has problems processing the anaphor when the subject skips that word?



Detect the **spillover** to the next word/words

```
RN: {'x1': -39.0, 'y1': -173.0, 'x2': 231.0, 'y2': -120.0, 'width': 270, 'height': 53, 'line': 2}
AN: {'x1': -315.0, 'y1': 61.0, 'x2': -183.0, 'y2': 114.0, 'width': 132, 'height': 53, 'line': 4}
SPILLOVER: {'x1': -183.0, 'y1': 61.0, 'x2': -51.0, 'y2': 114.0, 'width': 132, 'height': 53, 'line': 4}
```

Auf einer Konferenz zu Bildungsfragen wird derzeit über aktuelle schulische Themen gesprochen. Die **Lehrer*innen** sollen mehr auf **Role Noun** Disziplin und Konsequenzen im Unterricht setzen, doch aus pädagogischen Gründen, lehnen diese **Frauen** **die Vor**schläge ab. Die kommenden **Anaphor Spillover** Entwicklungen sind weiterhin Gegenstand von Diskussionen.



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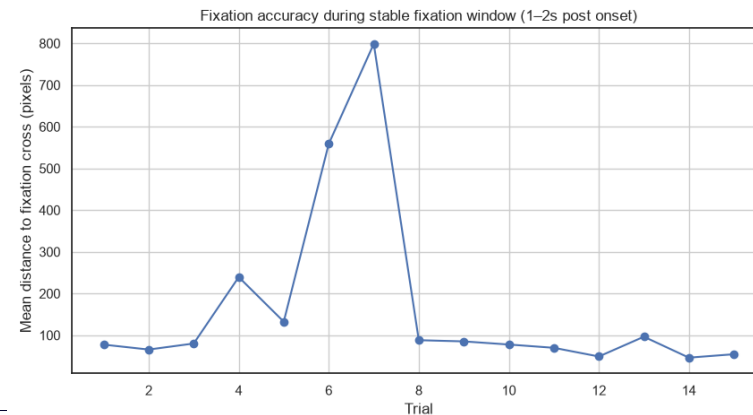
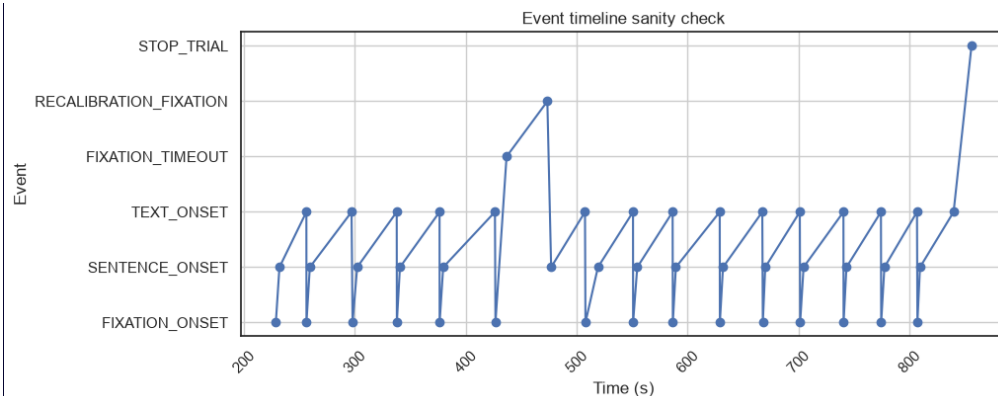
Quality Control

Planned Analysis

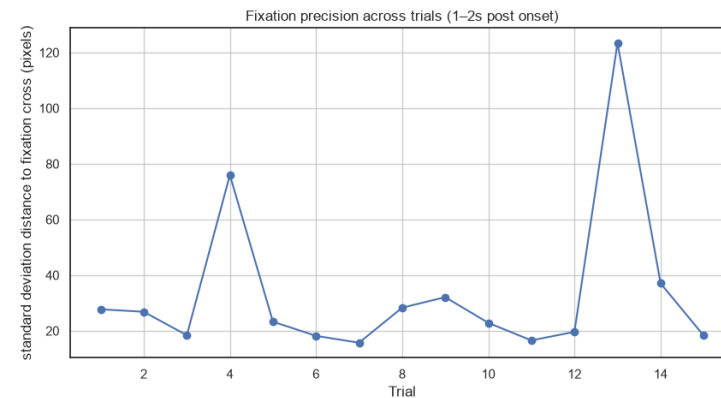
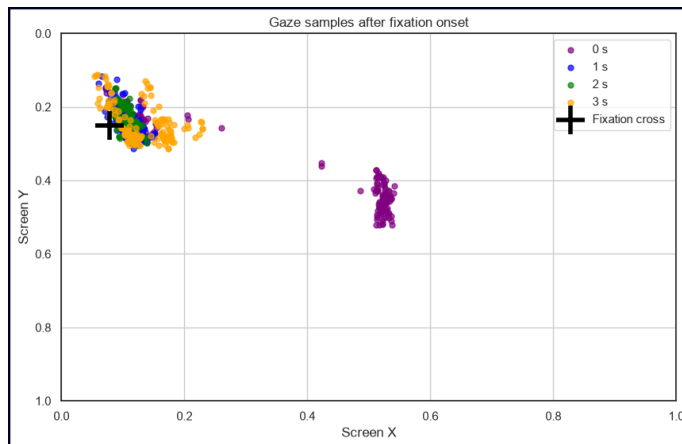
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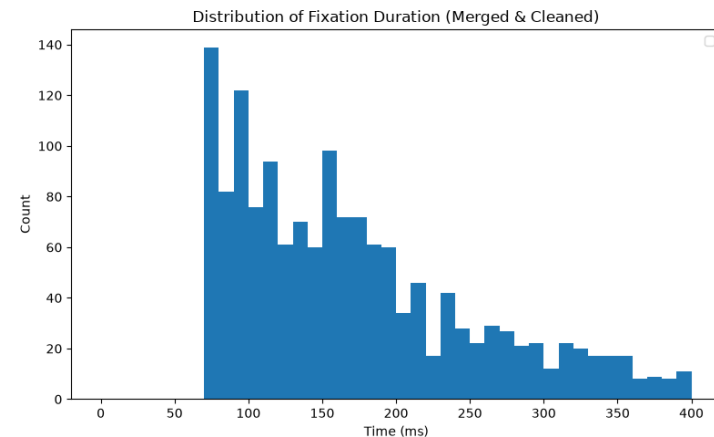
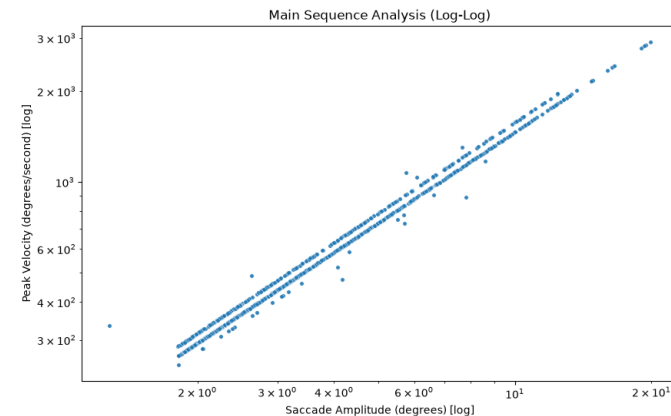
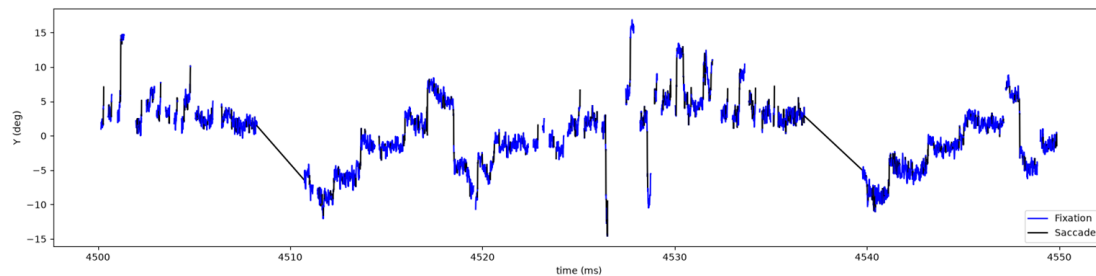
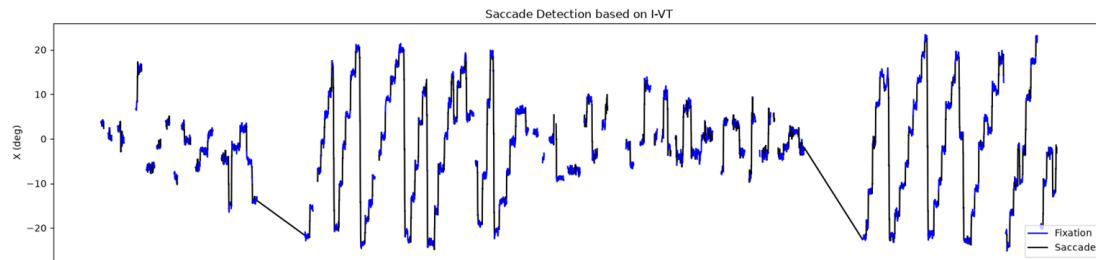
Accuracy & Precision



from pilot1
(b.f. the change
of design)

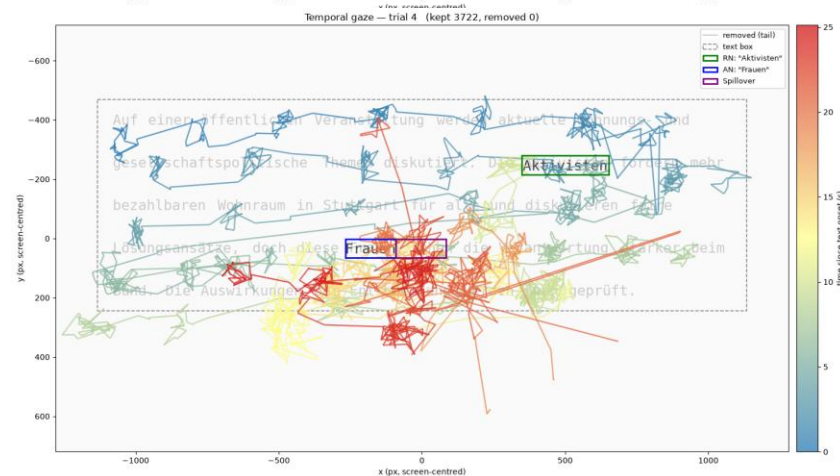
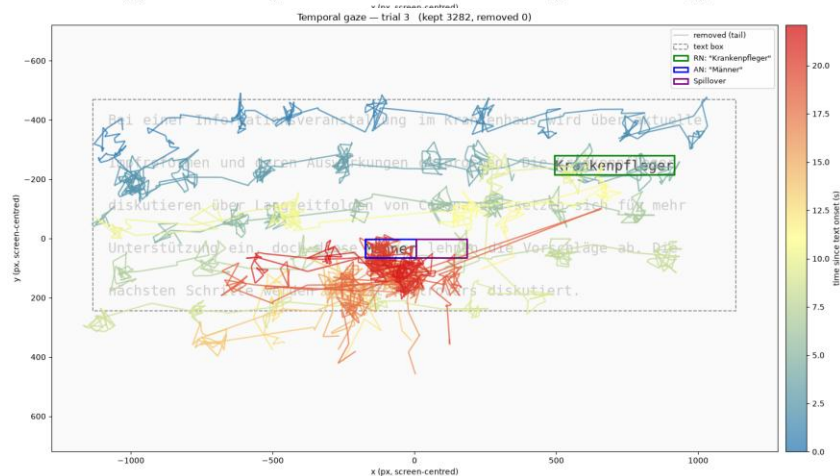
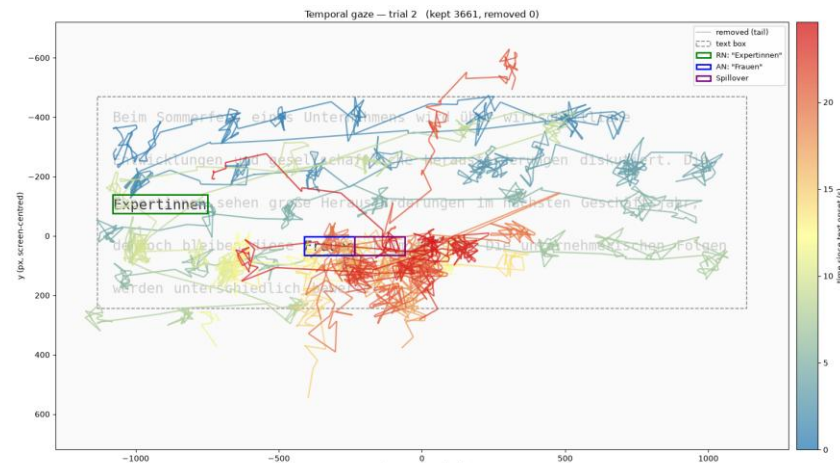
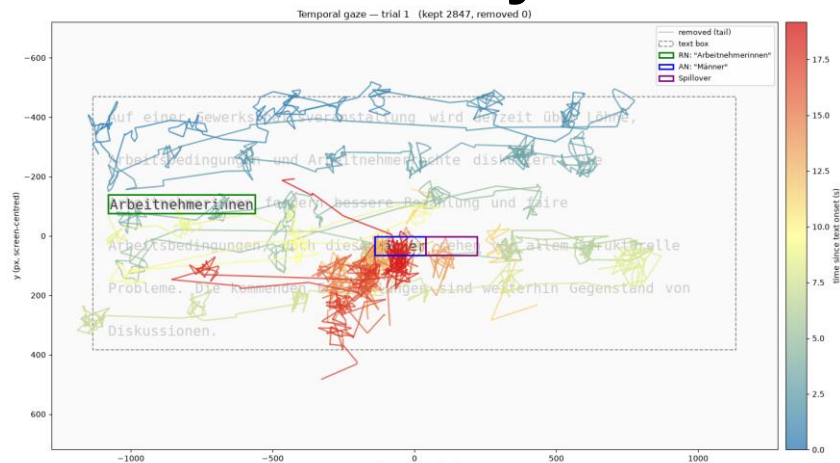


Fixation & Saccades



Gaze-text overlay

Quality Control





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Preprocessing



- Eye movements **recorded**
 - participants read 18 sentences
- **Exclude** invalid trials
 - Participant looked away
 - Participant did not read the whole sentence
- **Exclude** invalid data
 - remove blinks
 - track loss
- **Extract** Areas of Interest (AOIs)
 - Role noun
 - Anaphor
 - Spillover

Planned Analysis



- Extract eye-tracking **measures** for AOIs
 - Fixation Duration
 - Regression Probabilities
 - Go-past Time
- Compare eye-tracking measures **across conditions**
- **Analyze** questionnaire responses
- **Compare** questionnaire responses with eye-tracking measures and explore correlations

Expected Results

Condition	Expected Result
Masculine -> Männer	Fastest processing (gender match)
Masculine -> Frauen	Longer processing (possible male bias)
Feminine -> Frauen	Fastest processing (gender match)
Feminine -> Männer	Longest processing (gender mismatch)
Gender-inclusive -> Männer	Slightly longer processing (slight female bias)
Gender-inclusive -> Frauen	No strong gender bias expected (reduced bias)

Possible Outcomes

- **Significant** differences between conditions
 - > support for the hypotheses and **evidence of gender bias**
- **No significant** differences
 - > hypotheses are **not supported**
- **Correlations** between attitudes / experience / exposure and eye-tracking measures
 - > suggest that **individual factors influence** language processing



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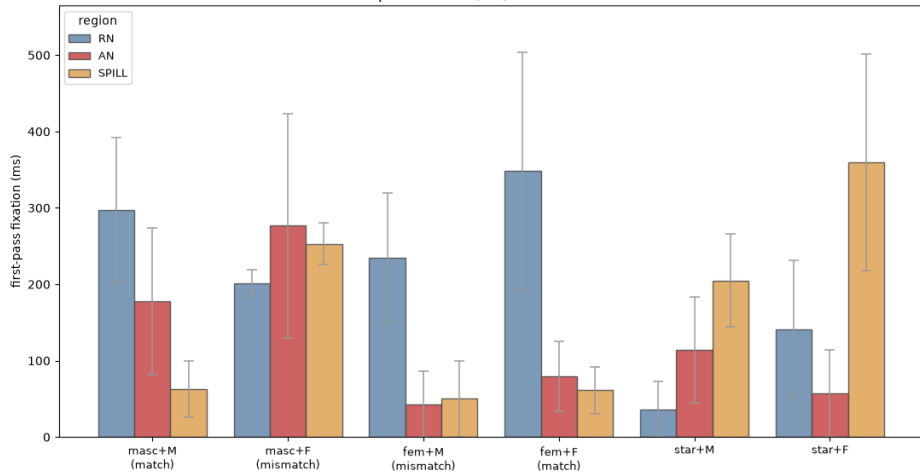
Analysis Results

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Fixation Duration

Analysis Results

first-pass fixation (ms) across conditions



First-pass fixation - AN

- masc+F > masc+M ✓
 - potential male bias
 - but masc+F being the highest? 🔍
- star+M > masc+F ✓
 - potential female bias
- fem+M is the lowest ⚡

Total fixation - AN

- large increase on fem+F ⚡

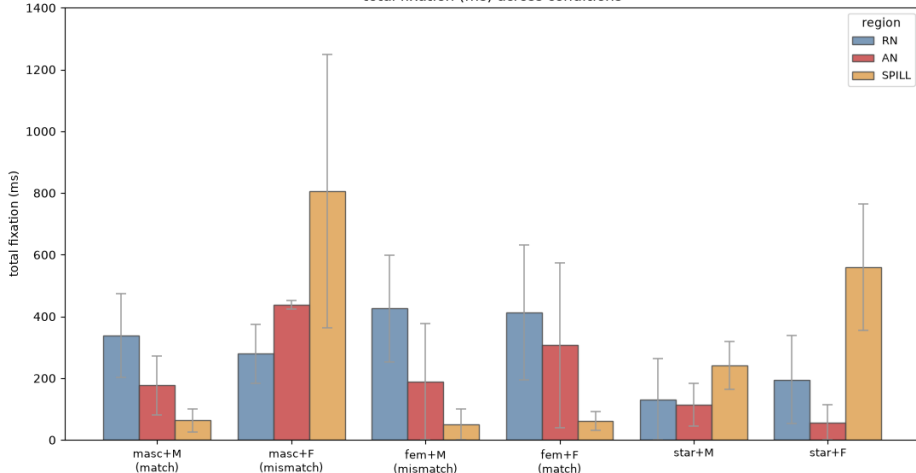
Difference between the pattern seen in first-pass and total fixation results

- Reflect the difference in early processing difficulty v.s. later integration

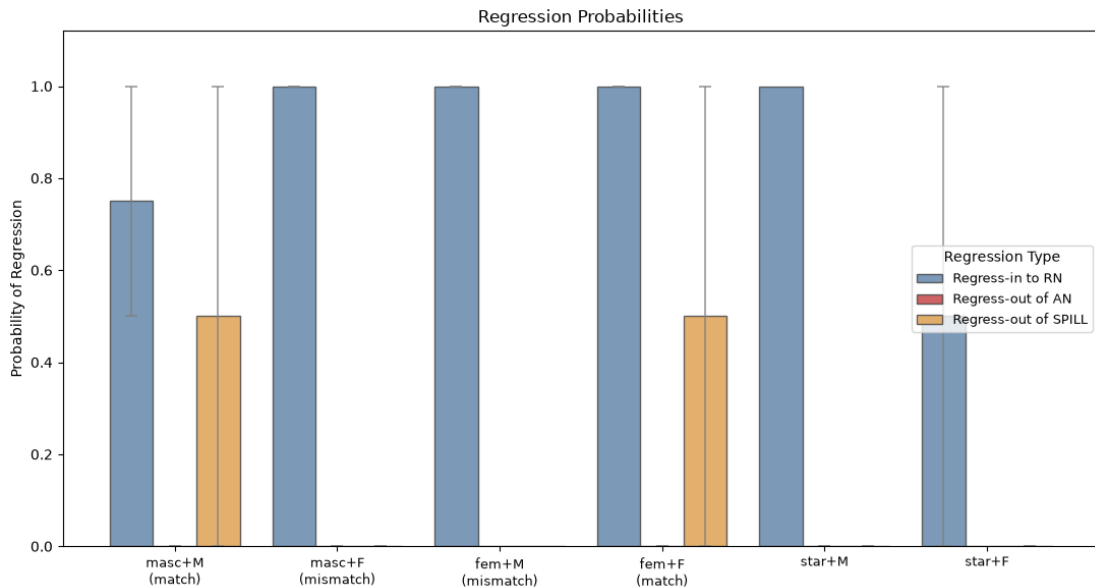
First-pass fixation - spillover

- much bigger after pronouns following the star form 🔍

total fixation (ms) across conditions



Regression Probabilities

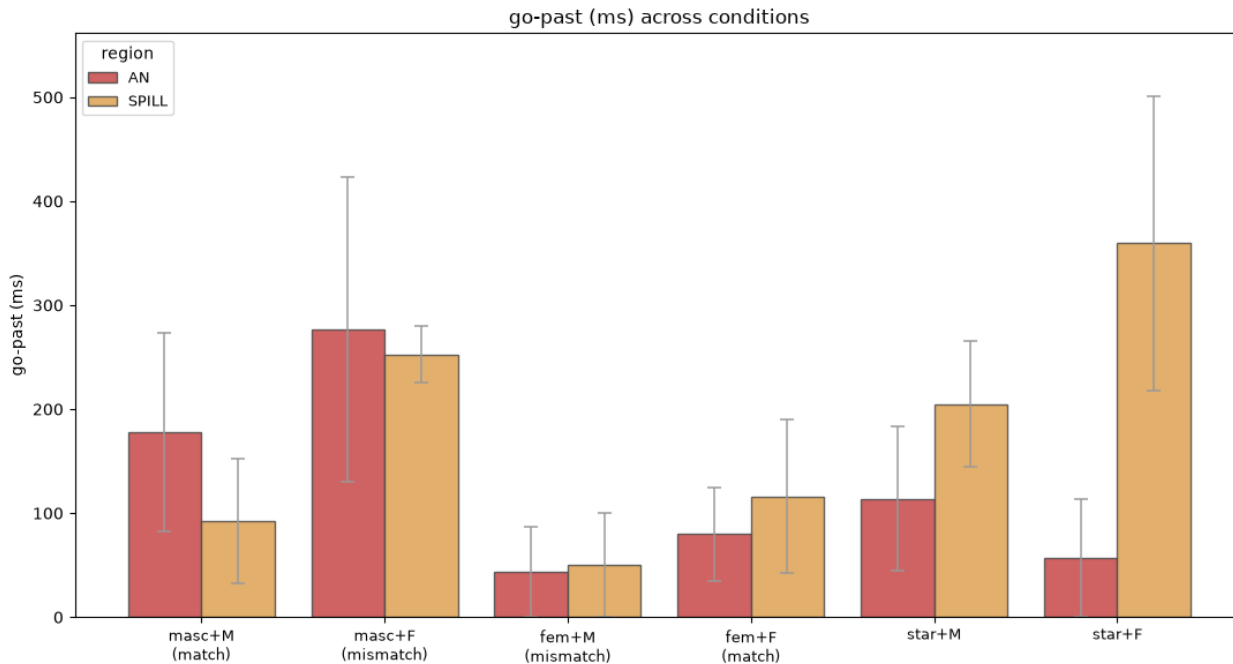


Regression-in on RN is frequent

- masc+F > masc+M ✓
 - potential male-bias
- star+M > star+F ✓
 - potential female-bias
- fem+F also very high 🔍
 - expected to be one of the lowest

Barely any regression-out

Go-past time



almost the same as the first-pass fixation •

note: the participant read rather fast and almost never went back until they were at the end

Questionnaire

Analysis Results

Post Experiment Questions

1 (low) – 5 (high) 1 2 3 4 5 not specified

1. How often do you use gender inclusive language:

In spoken language	☐	☒	☐	☐	☐	☐
In written language	☐	☐	☒	☐	☐	☐

- From one participant: **difference between written and spoken language exposure and usage**
- written -> fairly frequent (required in workspace)
 - spoken -> less frequent

2. How often do you notice gender inclusive language:

In spoken language	☐	☒	☐	☐	☐	☐
In written language	☐	☐	☒	☐	☐	☐

3. What is your opinion/standing towards gender inclusive language?

The participant has a positive attitude towards gender-inclusive language.

4. Where would you place yourself on a political spectrum from left to right?

Far left	Left of center	Center	Right of center	Far right	No answer
☐	☒	☐	☐	☐	☐

5. When did you first learn about gender-inclusive language?

Probably in school

Notes:



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Beyond eye-tracking measures

- captures implicit, physiological processing that self-report alone misses.
- Gender bias may operate below conscious awareness

Pilot Findings

- Regression pattern: gender inclusive plus female < masculine plus male < all other conditions
-> inclusive forms might be advantageous for including women

 Small pilot sample means results are tentative; the full study's larger sample is needed to confirm statistical robustness



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Thank you!

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